

EMPLOYEE RECORD MANAGEMENT SYSTEM

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Introduction

The Employee Record Management System is a Python-based application used to manage employee information. It allows users to add employee records, display all employees, search for an employee, and update employee details. This project is developed using Object-Oriented Programming (OOP) concepts to make the code modular, reusable, and easy to maintain.



Objectives

- Add employee records.
- Display employee details.
- Search employee by Employee ID.
- Update employee information.
- Apply Object-Oriented Programming concepts.
- Develop a simple menu-driven application.



Problem Statement

Managing employee records manually can be difficult and time-consuming. It may lead to errors and inefficient record management.

This project provides a simple solution to:

- Store employee information.
 - Search employee records.
 - Update employee details.
 - Display all employee information efficiently.
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Technologies Used

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1. Programming Language: Python
2. Concept: Object-Oriented Programming (OOP)
3. IDE: VS Code / IDLE / PyCharm
4. Platform: Windows



OOP Concepts Used

Abstraction

- Implemented using the Person abstract class.

Encapsulation

- Employee data is stored using private variables (__emp_id, __name, etc.).

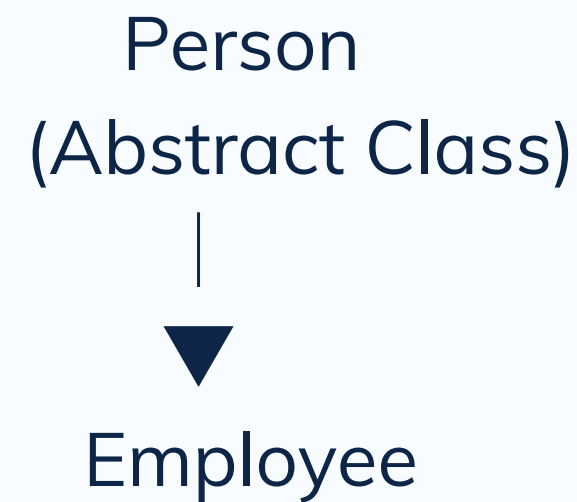
Inheritance

- Employee class inherits from the Person class.

Polymorphism

- The display() method is implemented in the Employee class.
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Class Diagram



Employee Class

Private Variables

- Employee ID
- Name
- Department
- Designation
- Experience
- Salary
- Bonus

Methods

- Constructor
- Getter Methods
- Setter Methods
- display()

System Architecture

L User



Main Menu

Add Employee	
Display Employee	
Search Employee	
Update Employee	
Exit	



Employee Object



Output

Features

- Add new employee records.
- Display all employees.
- Search employee by ID.
- Update salary, bonus, and experience.
- Calculate total salary.
- Menu-driven application.
- Uses Object-Oriented Programming concepts.

Result

Sample output

Employee Record Management System

1. Add Employee
2. Display All Employees
3. Search Employee
4. Update Employee
5. Exit

Display output

Employee ID : 101
Name : Prithika
Department : AI & DS
Designation : Data Analyst
Experience : 2 Years
Salary : 50000
Bonus : 5000
Total Salary: 55000

Advantages, Future Enhancements

Advantages

- Easy to use
- Simple menu-driven application
- Demonstrates all OOP concepts
- Reusable and maintainable code
- Calculates total salary automatically

Future Enhancements

- Database integration (MySQL)
- Login authentication
- Delete employee records
- Generate salary reports
- GUI using Tkinter or PyQt

Conclusion

The Employee Record Management System demonstrates the effective use of Abstraction, Encapsulation, Inheritance, and Polymorphism in Python. It provides a simple and efficient way to manage employee records and can be enhanced further by adding database support and a graphical user interface.



THANK YOU